Group Members: Rohan Shankar (rs2656), Harshaan Chugh (hsc53)
Paper Selection:
- Title: **LoRA: Low-Rank Adaptation of Large Language Models**
- Authors: Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, Weizhu Chen
- Publication Venue: International Conference on Learning Representations 2022
- Summary: LoRA proposes a more efficient way to fine-tune large pretrained language models without updating all their parameters. Normally, fine-tuning requires modifying millions (or billions) of weights, which is computationally expensive and memory-intensive. The authors see that the changes made during fine-tuning tend to be low-rank, meaning they can be approximated using much smaller matrices. So instead of updating the full weight matrices, LoRA freezes the original pretrained weights and adds small trainable low-rank matrices to certain layers. This reduces the number of trainable parameters while maintaining performance comparable to full fine-tuning. The paper shows that LoRA achieves similar or better results on benchmarks like GLUE and other language modeling tasks, while using far less memory and storage making it a really practical approach.
- Justification: We chose this paper because of its focus on making LLMs more efficient, which both of us thought were relevant. In addition, the idea behind LoRA of using smaller matrices to approximate larger ones was a cool alternative neither of us had considered, especially compared to other methods (such as optimizing the actual prompt you send into the LM).

Data:
- The two main benchmarks from the paper (GLUE, and WikiText-2) are publicly available:
    - https://gluebenchmark.com/
    - https://huggingface.co/datasets/mindchain/wikitext2

Result Selection:
- We want to reimplement the results on both of the datasets just mentioned. We plan to replicate the SST-2 and MRPC tasks from the GLUE benchmark, and compare the accuracy and F1 score between the baseline and LoRA versions across different ranks. We also will compare the model efficiency by reporting how many trainable parameters there were, and the amount of memory used (with LoRA being lower on both). We also want to replicate the language modeling results on WikiText-2 for measuring "perplexity."

Re-implementation Plan
- We will use BERT-base for the GLUE tasks, and GPT-2-small for the WikiText-2 tasks.
- For each of the attention layers, just as in the paper, we'll inject LoRA into some selected matrices, train them, and compare that to the full fine tuning without LoRA. We'll use all the same optimizers and other hyperparameters as in the paper, like AdamW.
- For metrics, we'll measure Accuracy and F1 score for GLUE, perplexity for WikiText-2, and for memory efficiency we'll measure # of trainable parameters, and the GPU memory consumption.
- We'll need ~8-10 GB, which should be easily covered by the Colab subscription
- We estimate we'll need around 6-10 hours of actual time running the experiments, which is also more than covered by the Colab subscription.